



HAOPENG (KEVIN) GENG

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Q ABOUT ME

Haopeng (Kevin) Geng is a Ph.D. candidate at the University of Tokyo whose research interests lie in **Human-Centered Speech AI**. His work focuses on developing acoustic and speech representations that capture speech nuances beyond mere word-level transcription, with applications to L2, accented, and pathological speech. From 2022 to 2024, he worked full-time as an AI Software Engineer at Laronix, engineering personalized speech recognition, voice conversion, and assessment technologies for people with voice loss. He is motivated by translating rigorous speech modeling into inclusive, real-world applications that empower diverse and underrepresented voices.

EDUCATION

The University of Tokyo, Tokyo, Japan

Ph.D. Candidate, Electrical Engineering and Information Systems

Apr 2024 – Mar 2027 (Exp.)

Advisor: Prof. Nobuaki Minematsu

Doctoral Research: Fine-Grained Acoustic Modeling, Phonetic Foundation Models, and Spoken-Language Assessment

Nagoya University, Aichi, Japan

M.S. in Informatics (Intelligent Systems)

Apr 2020 – Mar 2022

Advisor: Prof. Tomoki Toda

Thesis: Speech Conversion and Inpainting for Editing Disfluencies in Spontaneous Speech

Dalian University of Technology, Liaoning, China

B.S. in Computer Science and Technology & B.A. in Japanese (Double Degree)

Sep 2014 – Jun 2019

Undergraduate Exchange Program at Nagoya University

Sep 2017 – Aug 2018

PUBLICATIONS

- **Haopeng Geng**, Jiun-Ting Li, Daisuke Saito, Nobuaki Minematsu. Subphonetic Acoustic Modeling via Optimal Transport for Pronunciation Assessment, **SLT 2026**. [Paper]
- **Haopeng Geng**, Longfei Yang, Xi Chen, *et al.*, Beyond Acoustic Sparsity and Linguistic Bias: A Prompt-Free Paradigm for Mispronunciation Detection and Diagnosis, **INTERSPEECH 2026 Long Paper**. [[2nd Place, Iqra'Eval2 Challenge 2026]] [Paper] [Code]
- **Haopeng Geng**, Daisuke Saito, Nobuaki Minematsu, IF-MDD: Indirect Fusion for Prompt-Free Mispronunciation Detection and Diagnosis, arXiv preprint, 2026. [Paper] [Demo] [Code]
- **Haopeng Geng**, Daisuke Saito, Nobuaki Minematsu, A Perception-Based L2 Speech Intelligibility Indicator: Leveraging a Rater's Shadowing and Sequence-to-Sequence Voice Conversion, **INTERSPEECH 2025**. [Paper] [Demo]
- Ryoga Yamanaka, Keito Osa, Ayaka Fujiwara, **Haopeng Geng**, *et al.*, Synthesizing True Golden Voices to Enhance Pronunciation Training for Individual Language Learners, **SLaTE 2025**. [Paper]
- **Haopeng Geng**, Daisuke Saito, Nobuaki Minematsu, Simulating Native Speaker Shadowing for Non-Native Speech Assessment with Latent Speech Representations, arXiv preprint, 2025. [Paper] [Demo]
- **Haopeng Geng**, Daisuke Saito, Nobuaki Minematsu, A Pilot Study of Applying Sequence-to-Sequence Voice Conversion to Evaluate the Intelligibility of L2 Speech Using a Native Speaker's Shadowings, **APSIPA ASC 2024**. [Paper] [Demo] [Code]

FULL-TIME WORK EXPERIENCE

Laronix Pty Ltd, Queensland, Australia

AI Software Engineer (Full-time; transitioned to part-time after Ph.D. enrollment)

Mar 2022 – Oct 2024

AI & MedTech

- Developed personalized **speech recognition and voice conversion systems** for people with voice loss using an air-driven artificial larynx, reducing recognition error rates from **>50% to 20%** through speaker-specific adaptation.
- Collected and curated **high-quality, speaker-specific speech data** with the voice-loss community, achieving **effective low-resource personalization** with only **1–2 hours of recordings per speaker**.
- Built an end-to-end **clinician-facing prototype** for assessing speech intelligibility and naturalness and tracking **longitudinal changes** across recording sessions.

RESEARCH & INDUSTRY INTERNSHIPS

CoeFont Co., Ltd., Tokyo, Japan

Technical Intern

Mar 2025 – Sep 2025

Streaming ASR/TTS

- Explored streaming ASR adaptation (Parakeet-TDT) and evaluated TTS and data-cleaning approaches (e.g., CosyVoice, Emelia-Pipe) for real-time Japanese speech applications.

Carriage Inc., Tokyo, Japan

Technical Intern

Jan 2025 – Sep 2025

Speech Assessment

- Collected speech data from 50 non-native Japanese speakers and collaborated with academic researchers on human proficiency assessment.
- Developed a speaker-independent, session-level holistic proficiency scoring system, achieving 98% accuracy within ± 1 CEFR level.

Nagoya University, Aichi, Japan

Technical Assistant

Jul 2022 – Dec 2023

Research Support

- Facilitated large-scale speech database collection and maintained high-performance research computing environments.

NTT Media Intelligence Labs, Kanagawa, Japan

Research Intern

Feb 2021 & Sep 2021

Emotion Recognition and Speaker Diarization

- Investigated emotion recognition and speaker diarization using self-supervised pretrained speech models.
- Evaluated system robustness on complex mixture datasets for multi-speaker and paralinguistic speech research.

♡ HONORS AND AWARDS

Iqra'Eval2 Challenge (2nd Place) , INTERSPEECH 2026	Feb 2026
Miyabi Supercomputer Resource Grant for Young Researchers , UTokyo ITC	Mar 2025 – Present
SPRING GX Scholarship , The University of Tokyo	Apr 2024 – Present
Real-World Data Circulation Leaders Program , Nagoya University	Apr 2020 – Mar 2022
Scholarship for Outstanding Undergraduate Students , China Scholarship Council (CSC)	Sep 2017 – Aug 2018

⚙️ SKILLS

End-to-End ASR: CTC, hybrid CTC/AED, RNN-T

Speech Representation & Acoustic Modeling: self-supervised learning (SSL), continuous and discrete speech representations (Speech Codec), optimal transport-based acoustic modeling, HMM-based acoustic modeling, forced alignment

LLM-based Speech Understanding: speech encoder–LLM architectures, supervised fine-tuning (SFT), LoRA-based adaptation, chain-of-thought prompting, reinforcement learning fundamentals

Speech Generation: Transformer-based voice conversion; familiarity with flow matching and diffusion models

Programming: Python, Bash, C/C++, MATLAB

Frameworks & Tools: PyTorch, SpeechBrain, ESPnet, k2/icefall, Kaldi, Hugging Face, Gradio, Weights & Biases

Compute & Infrastructure: High-Performance Computing (HPC), multi-GPU and multi-Node distributed training, vLLM, Azure VMs, OpenAI API, GitHub

🌐 LANGUAGES

English: Fluent (IELTS 6.5/9.0)

Japanese: Fluent (JLPT N1, 165/180)

Mandarin: Native